

■ Phase 2: Requirement Analysis

Data Flow Diagrams and User Stories

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Project Name:	BOOK STORE (BookNest)

Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 (Context Diagram) Description

The bookstore platform interacts with three external entities:

- 1. **Customer:** Sends signup, authentication, search queries, cart additions, and checkout shipping data. Receives search results, wishlist feedback, and order summary.
- 2. **Seller:** Sends listings creation forms and uploads cover art images via HTTP POST. Receives inventory status, order inbox logs, and dashboard metrics.
- 3. **Admin:** Sends account validation approvals and moderation deletion operations. Receives platform summary data, user lists, and logs.

DFD Level 1 (Process Breakdown) Description

- Process 1.0 (Auth Controller): Reads login/signup payloads, checks user credentials in MongoDB, and signs JWT session tokens.
- Process 2.0 (Book Catalog engine): Decodes search queries, applies filters (genre, min/max price), and queries MongoDB Books collection.
- Process 3.0 (Order Controller): Pulls cart items, decreases Book collection stock counters, and writes new order document to MongoDB Orders collection.
- Process 4.0 (Seller Dashboard engine): Processes seller books creation and routes files using Multer into the filesystem storage (/uploads).
- Process 5.0 (Admin Dashboard engine): Handles admin actions to query store summaries and delete book entries.

User Stories Matrix

Role	Epic	USN	User Story	Acceptance Criteria	Priority	Sprint
Customer	Registration	USN-1	As a new user, I can signup with my email/password, so that I can browse as an authenticated member.	Validates email, hashes password using bcrypt, sets default status 'approved'.	High	Sprint -1
Customer	Login	USN-2	As a user, I can login, so that my cart/wishlist syncs across sessions.	Returns JWT token, updates frontend AppContext state.	High	Sprint -1
Customer	Search	USN-3	As a shopper, I can search books and filter by genre/price, so I find specific books quickly.	Real-time query mapping, case-insensitive title matching, price range check.	High	Sprint -1

Customer	Cart Flow	USN-4	As a buyer, I can add books to wishlist/cart and edit quantities, so that I checkout multiple items.	Stores cart in frontend state, sums total price, manages stock limits.	Medium	Sprint -2
Customer	Checkout	USN-5	As a customer, I can checkout by inputting shipping info, so that I place a formal order.	Stores order details in Order schema, clears user cart, decrements stock.	High	Sprint -2
Seller	Inventory	USN-6	As a seller, I can upload books with cover art, so that they appear in search results.	Form validation, Multer files write to disk, maps sellerId to session.	High	Sprint -3
Admin	Governance	USN-7	As an admin, I can delete any book listing and block users, to maintain store policy.	Checks admin role on route, database removes document, pulls from wishlists.	Medium	Sprint -3